KAMIL AKHTAR

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EDUCATION

Georgia State University **Dec 2026**
Master of Science, Computer Science *Atlanta, GA*
• **GPA:** 3.94/4.00

Institute of Business Administration (IBA) **Sep 2020 - May 2024**
Bachelor of Science, Computer Science *Karachi, Pakistan*
• **GPA:** 3.54/4.00

EXPERIENCE

ITA Group **Jun 2026 - Present**
AI Creative Intelligence Intern *Atlanta, GA*
- Built digital experience applications in Lovable (TypeScript) for Home Depot's "Unmatched" and Product Walk events, coordinating with design teams to launch on schedule across a 50,000 sq ft venue.
- Designed an interactive digital floor-plan application mapping every stall, product zone, and exhibit, giving attendees self-serve wayfinding and letting the client approve layout before build-out.
- Developed a T-shirt picker application on Lovable with a cloud database, letting attendees browse styles, select sizes, and order digitally in place of paper sign-up sheets and manual collation.
- Shipped "Brief Kit," an LLM-powered internal tool adopted by a 20-person creative team that produces structured end-to-end briefs with research, deliverables, and creative directions in roughly 7 minutes; owned requirements, scoping, and adoption.
- Ran marketing and event analytics on client and event success rates, building the visualizations and reporting database the team now uses as its single source for performance tracking.

TReNDS Center **Aug 2025 - Aug 2026**
Graduate Research Assistant *Atlanta, GA*
- Processed and analyzed resting-state fMRI from 1,140+ subjects across a multi-site clinical dataset using Python and MATLAB, building reproducible pipelines that applied independent component analysis (ICA) in a sliding-window framework, which generated dynamic brain-network features that supported downstream statistical analyses.
- Reduced high-dimensional data (5,460 features per window) with two-stage PCA and tested clinical group differences using covariate-adjusted general linear models with FDR correction, confirming stability through 100-iteration subsampling and split-half reproducibility analyses.
- Scaled the pipeline to the full cohort with parallel workloads on an HPC cluster (SLURM), making analyses tractable that were impractical on a single machine; results are being written up as a first-author manuscript.

Crestview AI **May 2025 - Aug 2025**
AI Integration Intern *Miami, FL*
- Built end-to-end AI automation pipelines with Python, n8n, Supabase, and LLM APIs (GPT-4o-mini, Mistral), reducing client operational latency by approximately 70%.
- Developed AI-enabled internal tools using LLMs and semantic embeddings, including a resume analyzer and recommendation tools that replaced manual review in client-facing workflows.
- Integrated and standardized data from multiple third-party APIs into one schema for downstream dashboards and models, and built reusable automation templates that cut manual effort by approximately 50%.

PROJECTS

QuadRAG - Multimodal RAG for Video Question-Answering
- Built a training-free multimodal RAG system fusing four semantic indexes - CLIP image embeddings, transcribed audio, frame descriptions, and domain captions - for video question-answering, engineering the full FastAPI, Streamlit, and vector-database stack to reach 61.8% answer accuracy and a 9.5% gain in retrieval performance.

High-Frequency Spatial Feature Fusion with 3D CNN for Early-Stage Schizophrenia Classification **2026**
- Proposed a high-frequency spatial feature fusion approach pairing high-pass spatial filtering with a modified 3D CNN (3D Laplacian kernel initialization, dilated convolutions), improving early-stage schizophrenia classification by 5.44% over baseline to 87% peak accuracy with a compact 1.4M-parameter model.

Genomic Sequence Indexing & Performance Benchmarking **2026**
ALSER Research Lab (GSU)
- C++, AMD-Sponsored, ALSER Research Lab (GSU)
- Benchmarked genomic k-mer indexing algorithms in C++ (ntHash vs. Minimap2-style bit manipulation) on datasets up to 100M characters, showing ~107M hashes/sec and a 90× speedup over a Python baseline; co-authored the "MetaBit" poster at GSU's 4th Annual Graduate Research Conference (2026).

TECHNICAL SKILLS

Languages: Python, SQL, C++, MATLAB, JavaScript, Bash
Machine Learning & AI: Deep learning, convolutional neural networks (CNNs), 3D CNNs, large language models (LLMs), retrieval-augmented generation (RAG), semantic embeddings, agentic AI workflows, natural language processing (NLP), model integration (GPT-4o, Mistral, Llama, CLIP), dimensionality reduction (PCA, ICA)
Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SciPy, Hugging Face Transformers, FastAPI, Streamlit
Data & Analytics: Statistical modeling, hypothesis testing, general linear models (GLM), false discovery rate (FDR) correction, ETL and data pipelines, data cleaning, exploratory data analysis (EDA), data visualization, KPI reporting, marketing and event analytics
Databases & BI: PostgreSQL, Supabase, vector databases, Tableau, Power BI, Microsoft Excel
Tools & Infrastructure: Git, GitHub, REST APIs, n8n, high-performance computing (HPC), SLURM, Jupyter, Linux, parallel and distributed processing
Business & Delivery: Requirements gathering, stakeholder interviews and communication, product scoping and management, cross-functional collaboration, technical documentation, testing, workflow automation, internal tooling